

EV Sonic Interaction Design: Data Acquisition Protocol

Project: Synthetic Electric Vehicle Soundscape

Focus: Audio UX & Parametric Synthesis (Python - Max/MSP)

Objective: High-fidelity OBD-II Data Logging for RPM, Engine Load, and Velocity Mapping.

1. PRE-FLIGHT & CALIBRATION

- ☐ **System Integrity Check:** Validate Python script stability and OBD-II interface connection
- ☐ **Data Verification:** Execute a 2-minute "pilot run" to ensure CSV formatting and timestamp precision (millisecond accuracy)
[FILE NAME: 01_CAL_Pilot_Run_Check.csv]
- ☐ **Baseline Idle State:** Log 60 seconds of stationary "System Ready" state (Minimum RPM, Null Load) [FILE NAME: 02_CAL_Stationary_Idle.csv]

Steady-State Velocity Profiles (Cruising):

- ☐ **30 km/h** (Constant velocity on level pavement)
[FILE NAME: 03_CAL_Cruising_30kmh.csv]
 - ☐ **50 km/h** (Constant velocity on level pavement)
[FILE NAME: 04_CAL_Cruising_50kmh.csv]
 - ☐ **90 km/h** (Constant velocity on level pavement)
[FILE NAME: 05_CAL_Cruising_90kmh.csv]
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2. DYNAMIC TRANSIENT PROFILES

Linear Acceleration (Throttle Mapping):

- ☐ **25% Throttle** (Gradual urban takeoff) [FILE NAME: 06_DYN_Acc_25_Percent.csv]
- ☐ **50% Throttle** (Standard maneuvering) [FILE NAME:07_DYN_Acc_50_Percent.csv]
- ☐ **100% Throttle** (Wide Open Throttle / Performance testing)
[FILE NAME:08_DYN_Acc_100_Percent.csv]

Deceleration & Energy Recovery:

- ☐ **Passive Coast-down** (0% throttle to complete stop)
[FILE NAME: 09_DYN_Coast_Down.csv]

Directional Inversion:

- ☐ **Transition from Drive (D) to Reverse (R)** to monitor RPM reset behavior
[FILE NAME:10_DYN_Direction_D_to_R.csv]
 - ☐ **Low-Velocity Parking** Maneuver [FILE NAME:11_DYN_Parking_Maneuvers.csv]
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3. STRESS TESTING

Gradient/Incline Profiles:

- ☐ **Steep Uphill Climb** (Targeting High Load / Low Speed correlation)
[FILE NAME:12_STR_Uphill_Climb.csv]
- ☐ **Controlled Downhill Descent** (Targeting Negative Load / Velocity tracking) [FILE NAME:13_STR_Downhill_Descent.csv]
- ☐ **Highway max speed** test [FILE NAME:14_Highway_max_speed.csv]

Anomalous Load Spikes:

- ☐ **Sudden throttle inputs** to test data jitter for interpolation calibration in Max [FILE NAME:15_STR_Traction_Load_Spikes.csv]
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4. INTEGRATED OPERATIONAL LOGGING

- ☐ **Urban Cycle:** High-density stop-and-go patterns and micro-movements. [FILE NAME:16_ENV_Urban_Center.csv]
- ☐ **Suburban/Open Road Cycle:** Sustained curves and variable pitch-load interactions [FILE NAME:17_ENV_Suburban.csv]
- ☐ **System Power Cycle (Start/Stop):** Log the transition from "Vehicle Off" to "Ready-to-Drive" and back to "Power Down" to identify data handshake patterns and initialization spikes. [FILE NAME:18_CAL_Power_Cycle.csv]

LIST OF FILE NAMES:

- 01_CAL_Pilot_Run_Check
- 02_CAL_Stationary_Idle
- 03_CAL_Cruising_30kmh
- 04_CAL_Cruising_50kmh
- 05_CAL_Cruising_90kmh
- 06_DYN_Acc_25_Percent
- 07_DYN_Acc_50_Percent
- 08_DYN_Acc_100_Percent
- 09_DYN_Coast_Down
- 10_DYN_Direction_D_to_R
- 11_DYN_Parking_Maneuvers
- 12_STR_Uphill_Climb
- 13_STR_Downhill_Descent
- 14_Highway_max_speed
- 15_STR_Traction_Load_Spikes
- 16_ENV_Center
- 17_ENV_Suburban
- 18_CAL_Power_Cycle.csv